Hardware noise and SIC state-space reconstruction

Circuit simulations, geometry, and diagnostic limitations

`cirq_sic` research record

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Abstract

We compare the probability body inferred from noisy real SIC triple-product circuits with the true quantum body and random-state SIC tomography. The procedures estimate different objects once errors are present. With qubit preparation depolarization at Cirq parameter $p = 0.05$, triple reconstruction has 72.665% of the true volume, while tomography has 81.304%. Common coherent rotations leave triples unchanged despite changing labeled tomography outcomes. Strong attenuation can make the nominal triple body empty without any failure of physical quantum dynamics. We provide a tested library pipeline, 39 qubit scenarios, three four-dimensional checks, compilation and shot-budget comparisons, and saved counts and scripts. All data are simulated; no hardware was tested.

1 Two different experimental objects

Let Π_i be trace-one SIC projectors in dimension d, with $n = d^2$, $G_{jk} = \mathrm{Tr}(\Pi_j\Pi_k) = (d\delta_{jk} + 1)/(d + 1)$ and $T_{ijk} = \mathrm{Re}\,\mathrm{Tr}(\Pi_i\Pi_j\Pi_k)$. For ideal SIC outcome probabilities,

$$\rho(p) = (d + 1)\sum_i p_i\Pi_i - I, \qquad (A_i)_{jk} = (d + 1)T_{jik} - G_{jk}. \tag{1}$$

The quantum body is exactly

$$K_0 = \{p \in \Delta_n : A(p) = \sum_i p_i A_i \succeq 0\}. \tag{2}$$

For Hermitian $X = \sum_j x_j\Pi_j$, the quadratic form is $\mathrm{Tr}(X\rho X)$. Positivity of ρ implies its nonnegativity; a negative-eigenvalue eigenvector projector supplies a violating X in the converse direction. Real triples suffice. Moment bounds alone do not give a complete mixed-state positivity test in general dimension. Background on SIC probability geometry and its algebra is given in Refs. [1, 2].

Our $\widehat{K}_T$ substitutes measured, permutation-averaged real triples while keeping the *ideal* Gram matrix. This is an experimental diagnostic, not a fit or certificate of a physical SIC. It may be inconsistent or empty. Tomography instead produces probabilities $q_i(\rho)$ through preparation and measurement errors. Fixed memoryless circuits give an affine map of the input density operator, whose nominal ideal-frame inverse can be unphysical. Its image K_{tomo} need not equal $\widehat{K}_T$. All comparisons use the same nominal SIC outcome labels. Recalibrating a frame changes the coordinate convention and must be reported explicitly.

2 Circuit design and reproducibility

Real triples use three independently prepared SIC states, a control Hadamard, two controlled register swaps, and a final control Hadamard. The measured expectation is $P(0) - P(1)$. Tomography uses

pure-state preparation and a SIC Naimark measurement unitary. Cirq density-matrix simulation uses complex128 arithmetic [3]. The maintained implementation is `cirq_sic.noise`.

Five channel families are swept at $0.01, 0.05, 0.15$: depolarization, bit flips, phase flips, phase damping and amplitude damping. Each is applied once after preparation or after every logical circuit moment. Readout bit flips use $0.01, 0.05, 0.15, 0.20$. Controls include the ideal case, common $R_z(0.4)$ preparation rotation, and three faults localized to the control or first system register: 39 qubit cases altogether. Localized register cases are not silently mapped to tomography, whose register roles differ.

Cirq depolarization shrinks a Bloch vector by $\lambda = 1 - 4p/3$. Bit/phase flips shrink two axes by $1 - 2p$; phase damping shrinks transverse axes by $\sqrt{1-\gamma}$. Amplitude damping has $M = \mathrm{diag}(\sqrt{1-\gamma}, \sqrt{1-\gamma}, 1-\gamma)$, $t = (0, 0, \gamma)$. Equal channel parameters are not equal infidelities or physical durations. Rotations are specified in radians.

Preparation noise acts only on system qubits, after preparation. An independent Kraus test caught that Cirq can pack the control Hadamard into the preparation moment; we explicitly split this moment to insert anisotropic noise correctly. Moment noise includes idle qubits and final basis rotations. The main sweep uses logical dense gates and CSWAPs; rates are per moment of that schedule, not native error rates. A supplementary CZ-gateset compilation has unrestricted connectivity and no device routing or duration calibration.

For $d = 2$ all 64 ordered settings are retained before symmetrization, at 4096 shots per setting. Tomography uses 384 Haar pure states and 4096 shots per state. Exact channel expectations and sampled counts are both stored. Binomial and multinomial draws from exact circuit probabilities reproduce terminal measurement statistics for these fixed, memoryless circuits. They do not model drift or correlated shots. The seed is 20261018. The recorded environment is Python 3.14.2, NumPy 2.3.5, SciPy 1.16.3 and Cirq 1.6.1.

The default tomography budget is six times the triple budget; no efficiency advantage is claimed from this comparison. Supplementary data use 682 tomography shots per state, or 261888 total, versus 262144 triple shots. Equal shots still do not imply equal gate, qubit, timing or calibration cost. For selected cases we generate 100 parametric binomial bootstrap tensors. Each independent setting is resampled once before permutation averaging. Endpoint empirical probabilities remain endpoints; percentile ranges are conditional sampling summaries, not calibrated coverage guarantees.

3 Metrics and explicit failure handling

Tensor consistency. Metrics include RMSE and maximum error against truth, raw permutation RMS, $T_{iii} = 1$ residuals, $T_{iij} = G_{ij}$ residuals, and closure violations $\sum_i T_{ijk} - dG_{jk}$. They test ideal SIC identities, not just precision. Symmetry-compressed data cannot reveal unmeasured order violations.

Full-dimensional geometry. From $p_* = \mathbf{1}/d^2$ we sample uniform directions in the sum-zero hyperplane, using 4096 common directions for qubits. Radius is limited by either the PSD boundary or the simplex. Whitening the center pencil reduces the PSD boundary to a minimum-eigenvalue calculation. Distances are Euclidean in probability coordinates. We report radial RMSE, sampled maximum radial discrepancy, radius ratio extrema, relative volume and Jaccard overlap. The maximum radial discrepancy is not Hausdorff distance. With $m = d^2 - 1$,

$$\frac{V_T}{V_0} = \frac{\mathbb{E} r_T^m}{\mathbb{E} r_0^m}, \qquad J = \frac{\mathbb{E} \min(r_T, r_0)^m}{\mathbb{E} \max(r_T, r_0)^m}. \tag{3}$$

A paired delta-method Monte Carlo standard error accompanies volume ratios; it excludes shot uncertainty. High-dimensional powers of radii can make Monte Carlo convergence slow, requiring more directions and repeated seeds.

If either center is not strictly feasible, radial metrics are withheld. This does not in general prove emptiness. A sufficient separate certificate is $\max_i \operatorname{Tr} A_i < 0$: every simplex combination then has negative trace. This certificate is incomplete. No general SDP feasibility search or adaptive recentering is implemented.

Tomography. We report paired total variation, empirical centroid shift, sorted covariance axis ratios, triple-pencil acceptance fraction and nominal ideal-inversion physicality. No positivity projection conceals unphysical inversions. A full-rank affine fit to known ideal input probabilities supplies singular values, center translation and residual; for qubits its determinant estimates the affine-ball volume. A small residual neither proves complete positivity nor distinguishes preparation and measurement errors. Covariance ratios also depend on input coverage, unlike singular values of a correctly specified fit.

Qubit hull volume is compared with the true ball and the ideal hull of the same states. Our ideal 384-state hull covers 96.896% of the true volume, illustrating finite-coverage bias. Shot noise can inflate hulls or create unphysical inversions. Beyond qubits, pure states are not the whole mixed-state boundary; a pure-state cloud is not a uniformly sampled full boundary.

4 Analytic mechanisms and counterexamples

4.1 Preparation depolarization

For tetrahedral Bloch vectors n_i, write $\rho_i = (I + \lambda n_i \cdot \sigma)/2$. Then

$$T'_{ijk} = \lambda^2 T_{ijk} + (1 - \lambda^2)/4. \tag{4}$$

In coefficient coordinates separating the uniform direction and its orthogonal complement, the nominal real pencil has a scalar block 1, a traceless block $(\lambda^2 - 2/3)I_3$, and off-diagonal coupling scaled by λ^2. Its inferred Bloch radius relative to truth is

$$R_T = \frac{\sqrt{3\lambda^2 - 2}}{\lambda^2}, \qquad R_{\text{tomo}} = \lambda \quad (\lambda > 0,\ \lambda^2 > 2/3). \tag{5}$$

At $p = 0.05$, volumes are 0.726648 and 0.813037; at $p = 0.10$, they are 0.300903 and 0.650963. For $\lambda^2 < 2/3$, the constant traceless principal block is negative, so the nominal triple body is empty. At equality it collapses to the center. The small-positive-p threshold is $p = \frac{3}{4}(1 - \sqrt{2/3}) \simeq 0.13763$. The channel and its tomography image remain physical. The inconsistency arises from applying ideal-SIC identities to mixed, non-SIC preparations.

4.2 Readout attenuation

A control bit flip with probability e gives $T' = sT$, $s = 1 - 2e$. In any dimension,

$$A_s(p) = sA(p) + (s - 1)G, \qquad \alpha = \frac{(d + 1)s - d}{s}. \tag{6}$$

For $d/(d + 1) < s \leq 1$, the inferred body is a contraction by α about p_*, with volume ratio α^{d^2-1}. For $s < d/(d + 1)$ every pencil has negative trace, certifying emptiness. The qubit threshold is $e = 1/6$. At $e = 0.05$ the triple volume is 0.470508. Independent readout flips on the two SIC output bits instead give nonuniform confusion eigenvalues s, s, s^2, so tomography has volume $s^4 = 0.6561$. These are different circuit/readout responses, not different quantum kinematics. Independently calibrated attenuation could be divided out, amplifying variance by $1/s^2$. Repeated-index triples alone cannot safely calibrate it if preparation purity and cycle-test gates are also faulty.

4.3 Gauge blindness and non-identifiability

A common unitary conjugation leaves all triple products invariant. In the $R_z(0.4)$ experiment the tensor and body agree with truth to roundoff while paired tomography has mean TV error 0.08006. The full tomography body also remains the same ball: shape alone misses this rotation too. Known input labels or an external reference frame are necessary. For qubit Bloch vectors,

$$\operatorname{Re}\operatorname{Tr}(\rho_i\rho_j\rho_k) = \tfrac{1}{4}(1 + r_i \cdot r_j + r_j \cdot r_k + r_k \cdot r_i). \tag{7}$$

A preparation channel $r \mapsto Mr + t$ therefore enters through M^TM, M^Tt and t^Tt. Real triples cannot determine a common output orthogonal orientation; complex conjugation preserves their real parts as well. Ancilla contrast loss and readout attenuation may be confounded. Volume or anisotropy alone does not uniquely identify a noise mechanism.

5 Numerical findings

Table 1 gives exact-channel qubit results at parameter 0.05, plus ideal and coherent controls. Dashes mean an invalid center, not zero volume. V_{tomo} is the fitted affine-ball volume. Moment noise can violate permutation symmetry despite an identical channel on each qubit, because controlled swaps expose registers differently.

Channel	Placement	V_T/V_0	V_{tomo}/V_0	TV	ϵ_{perm}
ideal	none	1.000	1.000	0.000	0.0000
depolarizing	preparation	0.727	0.813	0.017	0.0000
depolarizing	moment all	–	0.562	0.044	0.0100
bit flip	preparation	0.652	0.810	0.021	0.0000
bit flip	moment all	–	0.461	0.059	0.0133
phase flip	preparation	0.657	0.810	0.020	0.0000
phase flip	moment all	–	0.810	0.020	0.0108
phase damping	preparation	0.942	0.950	0.005	0.0000
phase damping	moment all	0.498	0.950	0.005	0.0042
amplitude damping	preparation	0.855	0.903	0.014	0.0000
amplitude damping	moment all	0.551	0.735	0.046	0.0073
bit flip	readout	0.471	0.656	0.034	0.0000
coherent z	preparation	1.000	1.000	0.080	0.0000

Table 1: Selected qubit simulations. TV uses paired ideal/noisy probabilities; ϵ_{perm} is raw permutation RMS.

Amplitude damping recovers the expected tomography singular values and translation, with volume $(1-\gamma)^2$. At $\gamma = 0.05$ its triple volume is about 0.8554, versus 0.9025 for tomography. Bit and phase flips have equal physical determinant volume but differently oriented contraction. Their finite Monte Carlo triple-volume estimates should be assessed with sampling errors, not interpreted as a robust difference in total volume.

At readout error 0.15, exact triple volume is 0.002915; 13 of 100 bootstrap centers are invalid. At 0.20 all 100 are invalid. Ideal-case bootstrap volume percentiles are $[0.9643, 1.0168, 1.0588]$. These illustrate sensitivity of nonlinear geometry estimates, particularly near feasibility boundaries; they are not an interval-coverage validation.

For $d = 4$, each exact-channel check uses 816 unordered settings. Ideal recovery has maximum tensor error about 2×10^{-15}. A 1% control readout error gives volume ratio 0.27877, explained by $(0.9/0.98)^{15}$. Local preparation depolarization at 1% gives 0.14034 with directional Monte Carlo standard

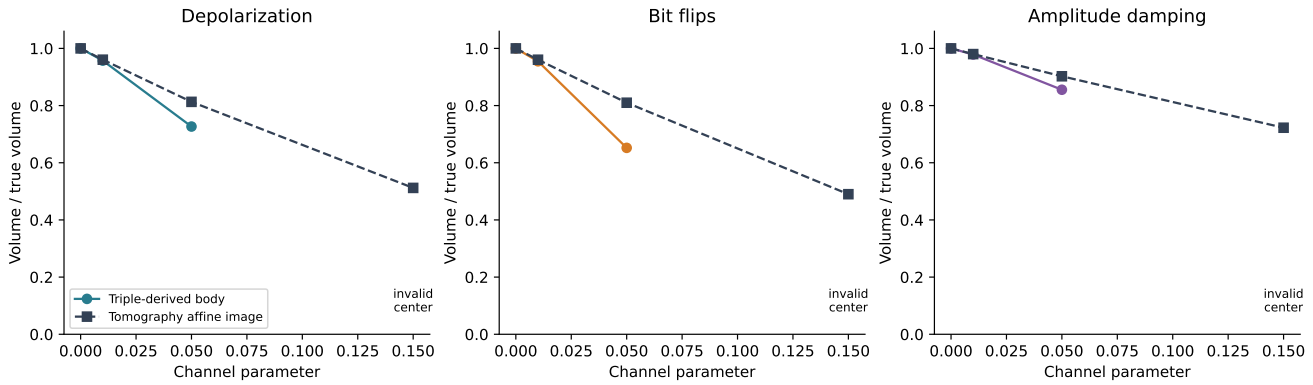


Figure 1: Preparation noise: triple and tomography volumes. Lines connect tested parameters. The invalid depolarizing center at 0.15 is analytically an empty body.

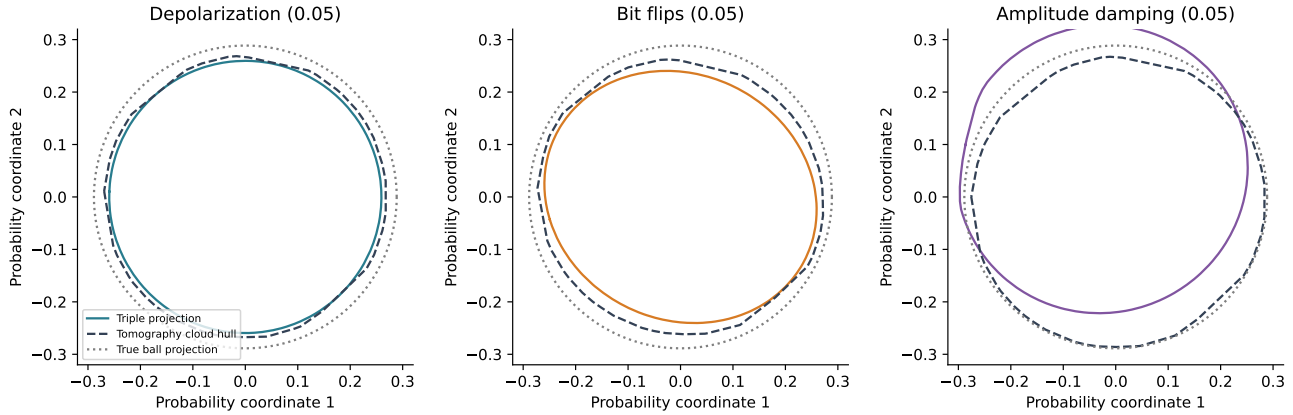


Figure 2: Two-coordinate projections, not sections, at parameter 0.05. Triple boundaries use 12000 radial endpoints; tomography uses a 384-state hull. Dotted circles show truth. Finite boundary/cloud sampling creates inward bias.

error 0.00542 on 2048 rays. This is exploratory, not a precision estimate. No $d = 4$ tomography cloud or ordered-asymmetry sweep is claimed.

Compilation to the CZ target gateset increases the qubit triple circuit from 4 to a mean 43 moments. With depolarization after each compiled moment, parameter 0.001 gives volume 0.36942; at 0.01 the body is certified empty by negative trace. These 20-setting symmetry-compressed checks demonstrate schedule sensitivity, not predictions for a specific chip. Native routing and durations remain to be modeled.

6 Practical hardware protocol

The method is useful as a structured consistency test and comparative benchmark:

1. Fix physical qubits, native compilation, routing, timing and nominal SIC labels. Record circuit depth and gate counts; interleave settings and repeats to separate drift from index dependence.

2. Acquire independent control contrast/readout and preparation purity or pair-overlap checks. Include repeated-index triples and deliberate register permutations before assuming tensor symmetry.

3. Compare raw identity and permutation residuals with shot uncertainty. Inspect feasibility before plotting. Publish failures and certificates beside geometry; do not smooth an inconsistent tensor into a plausible body.

4. Sweep depth, idle delay, register assignment and compiler settings. Use anisotropy, translation, attenuation and order violations as features; assign mechanisms only after interventions distinguish competing explanations.

5. Acquire labeled-state tomography under a recorded resource budget. Compare paired error, affine fits, nominal physicality and coverage, not only shape. Use held-out states or repeats to validate hardware model predictions.

6. For physical frame reconstruction, calibrate the Gram matrix and dual frame, or jointly fit a constrained gate/SPAM model. Report raw versus fitted geometry and model uncertainty. Such fitting is not implemented here.

7 Limitations and continuation

The current tools assume fixed memoryless channels and independent shots. They omit drift and leakage, and require custom models for crosstalk or correlated errors. Circuit helpers accept power-of-two dimensions and pure inputs, avoiding silent postselection of padded outcomes. Dense tensors cost $O(d^6)$ storage; ordered settings cost d^6, and density matrices scale exponentially in qubits. The default 4096-setting guard blocks accidental large studies.

Uniform-center failure needs SDP feasibility or recentering for general tensors. High-dimensional volume estimates need convergence studies. Near-singular PSD boundaries make bootstrap inference nonregular; repeated simulation experiments are needed to assess coverage. Beyond qubits, mixed-state ensembles or validated channel models are needed to infer the full body from tomography clouds. Priority next work is timing-aware native-device compilation, calibrated interventions, and ordered $d = 4$ data. Arbitrary Cirq noise models and compilation callbacks are already supported, so these can use the same analysis pipeline.

A Reproduction and validation

The study folder is `research/noise_state_space/`. Run `run_study.py`, `analyze_results.py`, and `supplementary.py` from the repository root; compile this document from its `report/` directory. NPZ files preserve states, settings, exact probabilities and counts; JSON files preserve metrics, failures, versions and seeds. The original prototype is archived as `pipeline.py`; maintained code is `cirq_sic/noise.py`. The public API and limitations are in `docs/noise_state_space.md`.

All 283 repository tests pass. New tests validate independent Kraus predictions for six channel/unitary families, analytic volumes, amplitude-damping affine parameters, coherent blindness, raw asymmetry, readout maps, shot/bootstrap reproducibility, compilation, invalid centers, singular clouds and resource guards. The regression run exposed a pre-existing persistence issue: repeated probability normalization changed values by an ulp on JSON reload. Already-normalized values are now preserved at machine precision, with a dedicated regression test. No hardware validation or confidence-interval coverage claim follows from these tests.

References

[1] D. M. Appleby, Å. Ericsson and C. A. Fuchs, *Properties of QBist State Spaces*, arXiv:0910.2750.

[2] D. M. Appleby, S. T. Flammia and C. A. Fuchs, *The Lie Algebraic Significance of Symmetric Informationally Complete Measurements*, arXiv:1001.0004.

[3] Google Quantum AI, *Noisy Simulation in Cirq*, https://quantumai.google/cirq/simulate/noisy_simulation, accessed 18 October 2026.